

# TAIGA-IACT: Parameters Definition Guide

Prepared by: Shahd Moghazy

START Programme Student

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# Introduction

This report is about describing the parameters used in the **TAIGA experimental** data.

TAIGA (*Tunka Advanced Instrument for cosmic ray physics and Gamma Astronomy*) experiment, located in Siberia, is a hybrid detector system designed to study high-energy gamma rays and cosmic rays. The main ones being the wide-angle air-Cherenkov array HiSCORE and the Imaging Atmospheric Cherenkov Telescopes (IACTs).

In this report we focus on the IACT technique, which records the Cherenkov light produced when high-energy particles interact in the Earth's atmosphere, creating extensive air showers (EAS). The IACT subsystem of TAIGA currently includes **three telescopes**, and the parameters analyzed in this report are taken from the experimental tables of the second telescope **TAIGA-IACT02**.

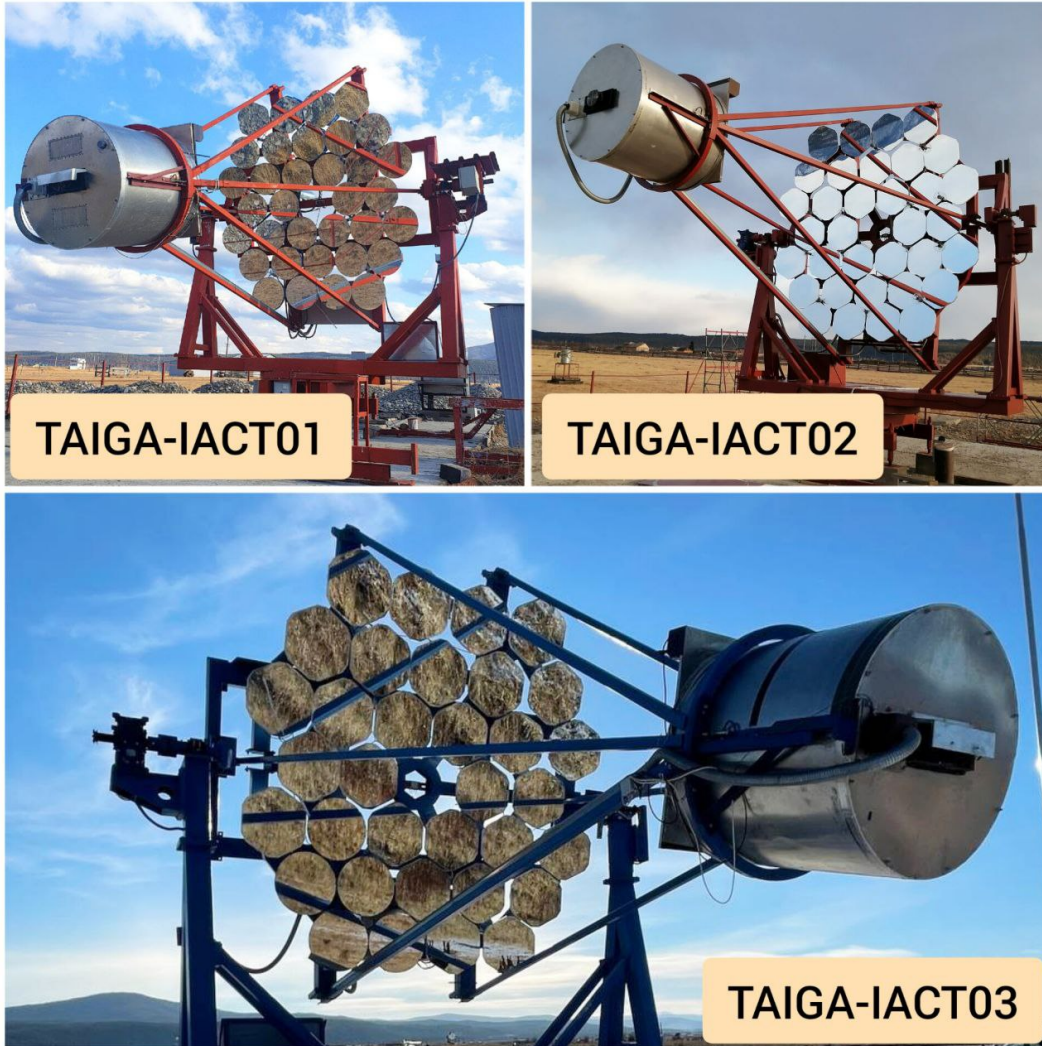


Figure 1: TAIGA IACT Telescopes

The dataset contains a wide variety of parameters. For better understanding, they will be grouped into several categories depending on their role in the event reconstruction and analysis.

Understanding their meaning and role are crucial, because not all of them carry direct physical meaning. Some describe the data structure, while others are important for extracting the physical properties of air showers and identifying gamma-ray events.

The categorization of parameters as follows:

1. Event Identification and Timing.
2. Telescope Geometry and Pointing.
3. Statistical/Hillas<sup>1</sup>.
4. Background Calibration.

In the following sections, each category will be discussed in detail.

## Parameters Definition

Understanding the characteristics of EAS requires determining a variety of parameters that describe both the geometric and physical properties of the detected events. These parameters help reconstruct the shower’s direction, energy, and morphology, allowing us to distinguish between gamma-ray and hadronic events. By analyzing them, we can evaluate the telescope’s performance, improve event selection criteria, and ultimately enhance the accuracy of astrophysical source identification.

To begin, we’ll examine the set of parameters and categorize them according to their common functions.

Table 1: List of parameters used in the dataset.

|             |            |                   |                |                     |
|-------------|------------|-------------------|----------------|---------------------|
| por         | event_num  | time              | unix_time_ns   | delta_time          |
| error_deg   | tel_az     | tel_el            | source_az      | source_el           |
| CR400phe    | CR_portion | median_background | MAD_background | numb_pix            |
| num_islands | size       | Xc                | Yc             | con_selected_island |
| con2        | length     | width             | dist1          | dist2               |
| a_axis      | skewness_w | skewness_l        | kurtosis_w     | kurtosis_l          |
| alpha1      | alpha2     | tel_ra            | tel_dec        | source_ra           |
| source_dec  | source_x   | source_y          | tracking       | good                |
| star        | edge       | weather_mark      | alpha_c        |                     |

### 2.1 Event Identification and Timing

These parameters are mainly used to label and organize events in the dataset. Although these parameters **do not** provide any physical information about the air shower itself, they form the first layer of metadata in the TAIGA-IACT dataset, enabling **identification**, **organization**, and **synchronization** of events across multiple telescopes.

Since this system consists of multiple telescopes, the synchronization of their recordings is crucial. They ensure that the same EAS observed by different telescopes can be correctly identified and combined into a single reconstructed event<sup>2</sup>. In this way, coincidence analysis<sup>3</sup> between telescopes becomes possible, improving the accuracy of direction and energy reconstruction.

<sup>1</sup>They are called Hillas parameters after A.M. Hillas, who introduced them in 1985 to describe the shape and orientation of Cherenkov light images from air showers.

<sup>2</sup>Meaning the shower information recorded by the telescope has been processed to estimate the properties of the original particle, such as its type, energy, and incoming direction.

<sup>3</sup>Coincidence analysis means matching events recorded by different telescopes that occur within a short time window, ensuring they belong to the same air shower rather than random background signals.

## What they are

1. **Portion number**(por)

It is used to segment the dataset into manageable chunks or files.

2. **Event number** (event\_num)

A unique identifier assigned to each detected shower. Helps distinguish events in the dataset and track them across processing steps.

3. **Timestamp** (time)

The time when the telescope recorded the event, in `hh:mm:ss:ms: $\mu$ s:ns4` format.

4. **Unix time** (unix\_time\_ns)

Timestamp expressed in nanoseconds since the Unix epoch (Jan 1, 1970). Useful for precise calculations, sorting events, and syncing between telescopes.

5. **Delta time** (delta\_time)

Represents the time difference between the current event and the previous event in the dataset from the same telescope (it's relative not absolute). It is typically expressed in seconds or fractions of a second. Indicates the event rate or detects gaps in recording.

6. **Error estimates** (error\_deg)

Indicates the telescope's pointing uncertainty in degrees, which reflects how accurately it was aimed at the source.

## 2.2 Telescope Geometry and Pointing

These parameters describe the **orientation of each telescope** and the **position of the observed source** in the sky. They are essential for reconstructing the direction of incoming EAS and for combining observations from multiple telescopes in the array.

Unlike event identification parameters, which are purely metadata, pointing parameters carry geometric information that links the telescope's orientation to the celestial coordinates of the observed source.

TAIGA-IACT provides three types of coordinate representations:

- **Horizontal coordinate (azimuth and elevation)**: describe the telescope pointing or the source position relative to the local horizon.  
These are **time-dependent** as the sky moves with respect to the observer.
- **Equatorial coordinate (declination and right ascension)**: describe the true position of the source. These are **independent** of the observer's location.
- **Camera coordinate (x,y)**: describe the projection of the source's position onto the telescope's camera plane. These are used for imaging analysis and event reconstruction in the detector.  
This system is defined relative to the telescope's optical axis, with the field of view (FoV) determining the angular coverage of the plane which is  $\sim 9^\circ$  for TAIGA-IACT.

A detailed explanation of the horizontal and equatorial coordinate systems, is provided in Appendix A.

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<sup>4</sup>This is hour: minute: seconds: milli-seconds: micro-seconds: nano-seconds

## What they are

1. **Telescope azimuth (`tel_az`)**

The azimuthal angle of the telescope, measured in degrees from North towards East.

2. **Telescope elevation (`tel_el`)**

The elevation angle of the telescope above the local horizon, measured in degrees.

3. **Telescope right ascension (`tel_ra`)**

The right ascension of the telescope pointing in equatorial coordinates, measured in degrees.

4. **Telescope declination (`tel_dec`)**

The declination of the telescope pointing in equatorial coordinates, measured in degrees.

5. **Source azimuth (`source_az`)**

The azimuth of the source in the horizontal coordinate system.

6. **Source elevation (`source_el`)**

The elevation of the source in the horizontal coordinate system.

7. **Source right ascension (`source_ra`)**

The right ascension of the source in equatorial coordinates.

8. **Source declination (`source_dec`)**

The declination of the source in equatorial coordinates.

9. **Source camera coordinates (`source_x`, `source_y`)**

The position of the source projected onto the telescope's camera plane.

Figures 2 illustrate the telescope orientations in the horizontal and equatorial coordinate systems, with the third figure illustrating how the source is projected onto the telescope's camera plane.

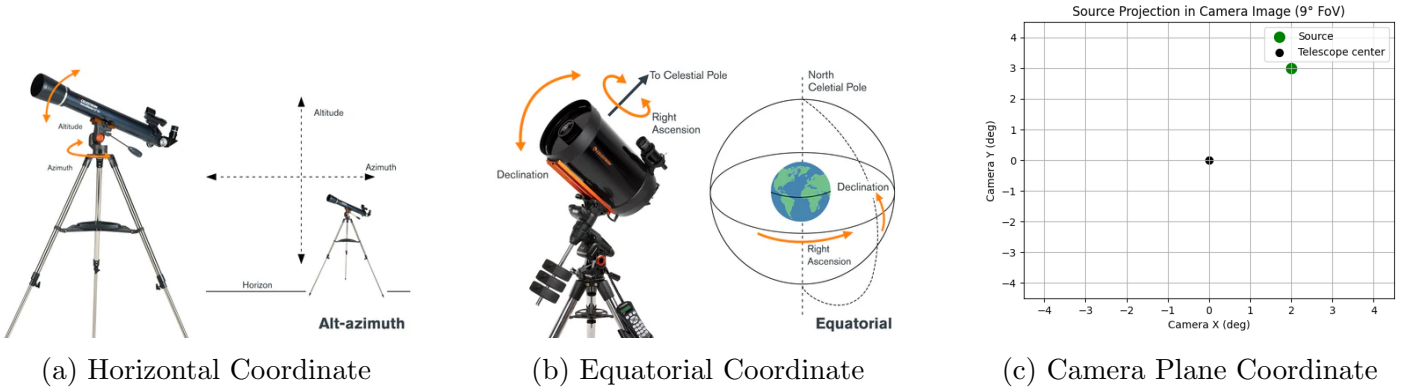


Figure 2: Three Coordinate Systems Used by TAIGA-IAC

The telescope's horizontal coordinates (`tel_az`, `tel_el`) describe the pointing direction relative to the local horizon at a given moment and are therefore time-dependent, changing continuously due to Earth's rotation, which makes the sky appear to move.

In contrast, the equatorial coordinates (`tel_ra`, `tel_dec`) define the telescope's absolute pointing on the celestial sphere and are independent of both the observer's location and the time of observation.

## Why do we need both?

Given the source position in RA/Dec and the telescope pointing in Az/El, we must convert<sup>a</sup> the coordinates to a common system in order to aim the telescope at the source.

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<sup>a</sup>Converting between horizontal and equatorial coordinates involves the observer's location and the local sidereal time(LST).

Together, these coordinate representations enable precise reconstruction of incoming air showers by correlating the source position with the telescope orientation and recorded images.

## 2.3 Hillas Parameters

The Hillas parameters are a set of **geometric** and **statistical** descriptions of the shower image ellipse. Hillas proposed a simple but powerful method to characterize Cherenkov images produced by EAS in IACTs. They are named after him because this parametrization was the *first systematic* way to reduce a complex image into a small number of descriptive quantities that capture the showers' orientation, shape, and size.

Originally, Hillas introduced a large set of parameters, but many of them carried overlapping information. Over time, IACTs analysis methods were refined, so some of these parameters were removed and others remained in common use.

But it did not stop there, modern IACTs analysis techniques have extended the original set by adding **new parameters** that account for higher-order image properties and background discrimination.

## What they are

### 2.3.1 Core Hillas Parameters

Core Hillas parameters are those that describe image size, centroid, shape and orientation.

1. **Size (size)**<sup>5</sup>  
The total number of photoelectrons (phe) in the image shows the brightness of the shower and is proportional to the primary particle's energy.
2. **Centroid coordinates (Xc, Yc)**  
The position of the image (shower) centroid (center of gravity) in the camera, calculated as the weighted average of pixel positions.
3. **Length (length)**  
The second-order moment along the major axis of the fitted ellipse. It represents the longitudinal development of the shower (how elongated the shower is).
4. **Width (width)**  
The second-order moment along the minor axis of the ellipse, describing the lateral spread of the shower in the camera.

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<sup>5</sup>The **size** parameter is calculated as a weighted mean rather than a simple mean. This gives more physical significance to the recorded signals, since pixels with higher amplitudes (those carrying more information about the shower structure) contribute more strongly to the total value.

5. **Slope of major axis (a\_axis)**

The slope coefficient of the ellipse's major axis relative to the camera grid  $y = ax + b$ . It characterizes the orientation of the shower image and represents the direction of the major axis in the camera plane.<sup>6</sup>

6. **Alpha angle (alpha1, alpha2)**

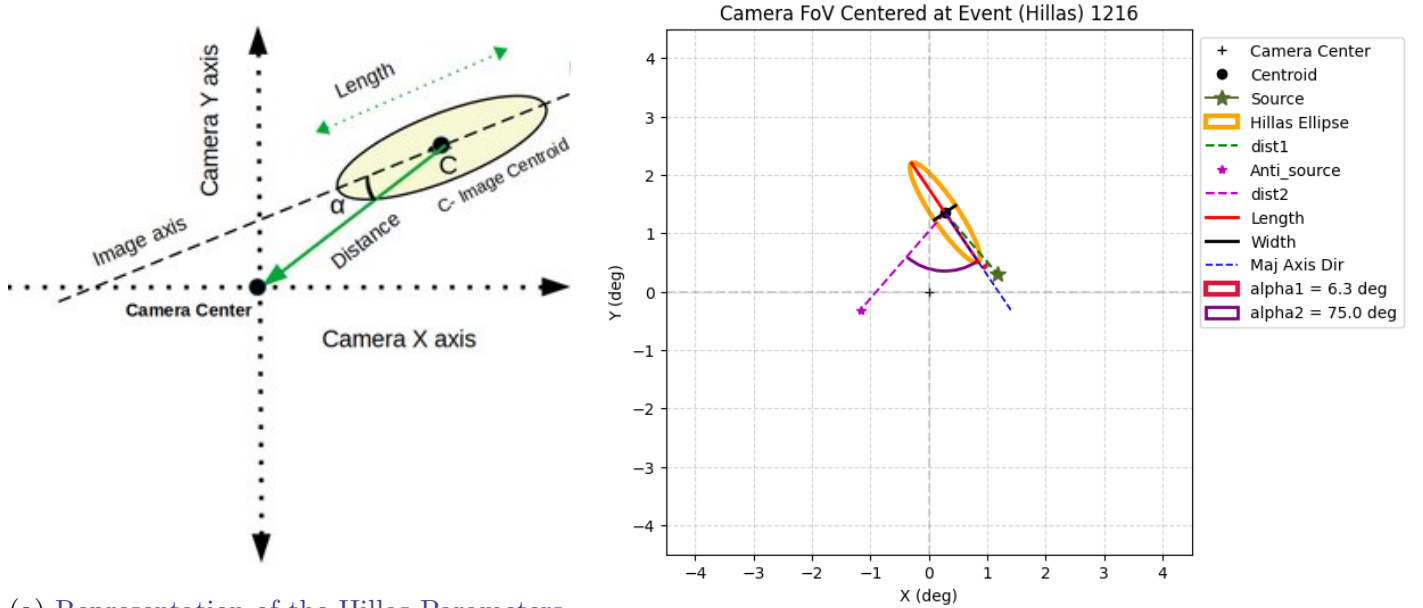
The angle between the ellipse's major axis and the source (or anti-source) direction. For gamma rays,  $\alpha$  is usually small, as the image points back to the source. While hadronic events, typically have larger  $\alpha$  values. **It is crucial for gamma/hadron separation.**

7. **Distance to source (dist1)**

The angular distance between the image centroid and the expected source position in the camera.

8. **Distance to anti-source (dist2)**

The angular distance between the image centroid and the anti-source position, defined as the point diametrically opposite to the source (through the camera center).



(a) Representation of the Hillas Parameters (Diagram)

(b) Real Event Representation from Our Data

Figure 3: Hillas Parameters

**Note:**

- The **a\_axis** and **alpha1** parameters together describe the shower orientation relative to the source.
- The parameter **dist** has two conventions, depending on the analysis framework used:
  1. Distance from the image centroid  $(x_c, y_c)$  to the **camera center**  $(0, 0)$ :

$$\text{dist} = \sqrt{(x_c - 0)^2 + (y_c - 0)^2}.$$

<sup>6</sup>For analysis and plotting, we can define an angle  $\phi$ , which is the  $\tan^{-1}(\text{a\_axis})$ .



2. Distance from the image centroid to the **expected source position**  $(x_s, y_s)$  in the camera:

$$\text{dist} = \sqrt{(x_c - x_s)^2 + (y_c - y_s)^2}.$$

But here we use the 2nd convention.

- There also another parameter, that often appear in IACT image: **Displacement (disp)** is defined as the distance from the image centroid to the source position *along the major axis of the Hillas ellipse*.

It is reconstructed from the measured **dist** by projecting it onto the main axis:

$$\vec{r}_{\text{source}} = \vec{r}_{\text{centroid}} \pm \text{disp} \hat{u}_{\text{major}},$$

where  $\hat{u}_{\text{major}}$  is the unit vector along the ellipse's major axis.

### 2.3.2 Higher-Order Shape Parameters

1. **Skewness** (`skewness_l`, `skewness_w`)

Third-order image moments describing the asymmetry of the light distribution along the major and minor axes. They provide information about whether the shower is tilted toward or away from the source, positive values indicate a shift toward the source.

2. **Kurtosis** (`kurtosis_l`, `kurtosis_w`)

Fourth-order moments measuring how peaked or flat the image distribution is compared to a Gaussian profile.

In summary, the **classical Hillas parameters** consist of *size*, *centroid* ( $X_c$ ,  $Y_c$ ), *length*, *width*, *orientation* ( $a_{\text{axis}}$ ,  $\alpha$ ), and *distances*, with optional higher-order moments (*skewness*, *kurtosis*).

For more explanation on the statistical definitions of these parameters see appendix B.

### 2.3.3 Modern Extension

These parameters are not part of the original Hillas definition, but added in modern analysis.

1. **Number of islands** (`num_islands`)

The number of isolated pixel clusters after cleaning. Gamma-ray showers are usually compact, while hadronic showers may produce several islands.

2. **Concentration in brightest part** (`con_selected_island`)

The fraction of the total light (*size*) contained in the brightest island after cleaning.

3. **Concentration in brightest pixels** (`con2`)

The fraction of the total light contained in the two brightest pixels, used as a measure of compactness.

4. **Number of cleaned pixels** (`numb_pix`)

The total number of pixels that passed the image cleaning procedure.



### 2.3.4 Additional metadata

Quality-control and observational parameters that provide information about the telescope status and observational atmospheric conditions:

- **tracking**: Indicates whether the telescope was actively tracking the source (1) or not (0).
- **good**: Flags if the telescope pointing was accurate and tracking errors were small (1 = good, 0 = not good).
- **star**: Indicates if a bright star was present among the cleaned pixels. Stars can affect pixel amplitudes; if detected.
- **edge**: It indicates whether the shower image lies partly on the edge of the camera or not. If it is flagged (means =1 or set to True), it warns that the event's shape parameters may be unreliable because part of the image could be missing.
- **weather\_mark**: Estimates the observation conditions using the CCD camera, typically ranging from 10 (excellent visibility) to -1 (no data available).
- **alpha\_c**: Camera rotation angle accounting for deviations from orthogonality of axes and telescope tilt, used to correct orientation measurements.

## 2.4 Background Calibration Parameters

The parameters related to background calibration are those that quantify the baseline signal in the camera and its fluctuations. They are crucial for calibrating the camera response and separating true gamma-ray events from background noise.

### What they are

#### 1. **median\_background**

It is the typical signal level from the camera's background noise, estimated from pixels that didn't contain Cherenkov light from the shower. It helps quantify how bright or noisy the sky was during that portion of data.

#### 2. **MAD\_background**

The Median Absolute Deviation (MAD) quantifies how much the background noise varies across non-cleaned pixels. It is a robust measure of the background's spread or fluctuation, complementing the `median_background` value.

### Event-rate Parameters

#### 1. **Count rate with size > 400 photoelectrons (CR400phe)**

The event count rate after image cleaning, restricted to events with a total size greater than 400 phe. The rate is averaged over 30 seconds of observation and is used as a stability and quality check of telescope performance.

#### 2. **Portion count rate (CR\_portion)**

The average count rate of all events that passed image cleaning, calculated over a selected observation interval (portion). Unlike `CR400phe`, this parameter is not restricted to high-size events and reflects the general event rate during the observation.

**Note:**

Parameters like `tracking`, `good`, `star`, `edge`, `weather_mark`, and `alpha_c`, describe the telescope status or observational conditions and are considered metadata.

But parameters as `CR400phe` and `CR_portion`, are quality control parameters. They summarize the event rate over a time interval rather than properties of individual events or the telescope, and therefore are treated separately from the metadata.

## 2.5 Summary for Parameters Description

| Parameter                 | Description                                                    |
|---------------------------|----------------------------------------------------------------|
| <code>por</code>          | Portion number.                                                |
| <code>event_numb</code>   | Used to track events number inside the portion.                |
| <code>time</code>         | Event timestamp in <code>hh:mm:ss:ms:μs:ns</code>              |
| <code>unix_time_ns</code> | Timestamp in <b>nanoseconds</b> since Unix epoch.              |
| <code>delta_time</code>   | Time difference from the previous event in the same telescope. |
| <code>error_deg</code>    | Telescope pointing uncertainty in <b>degrees</b> .             |

Table 2: Event identification parameters.

| Parameter                       | Description                                                       |
|---------------------------------|-------------------------------------------------------------------|
| <code>tel_az</code>             | Telescope azimuth in <b>degrees</b> .                             |
| <code>tel_el</code>             | Telescope elevation in <b>degrees</b> .                           |
| <code>tel_ra</code>             | Telescope right ascension in <b>degrees</b> .                     |
| <code>tel_dec</code>            | Telescope declination in <b>degrees</b> .                         |
| <code>source_az</code>          | Azimuth of the source in <b>degrees</b> .                         |
| <code>source_el</code>          | Elevation of the source in <b>degrees</b> .                       |
| <code>source_ra</code>          | Source right ascension in <b>degrees</b> .                        |
| <code>source_dec</code>         | Source declination in <b>degrees</b> .                            |
| <code>source_x, source_y</code> | Source position projected on the camera plane in <b>degrees</b> . |

Table 3: Telescope pointing parameters.

| Parameter                   | Description                                                                           |
|-----------------------------|---------------------------------------------------------------------------------------|
| <code>size</code>           | Total number of photoelectrons (brightness of shower) in <b>phe</b> .                 |
| <code>Xc, Yc</code>         | Centroid coordinates of the shower in <b>degrees</b> .                                |
| <code>length</code>         | Second-order moment along major axis.                                                 |
| <code>width</code>          | Second-order moment along minor axis .                                                |
| <code>a_axis</code>         | Slope of ellipse major axis relative to camera grid (orientation) in <b>degrees</b> . |
| <code>alpha1, alpha2</code> | Angle between major axis and source/antisource in <b>degrees</b> .                    |
| <code>dist1</code>          | Distance between centroid and source position in <b>degrees</b> .                     |
| <code>dist2</code>          | Distance between centroid and antisource position in <b>degrees</b> .                 |

Table 4: Hillas parameters describing shower images.

| Parameter              | Description                                                                |
|------------------------|----------------------------------------------------------------------------|
| skewness_l, skewness_w | Third-order moments describing asymmetry along major and minor axes.       |
| kurtosis_l, kurtosis_w | Fourth-order moments measuring peakiness or flatness compared to Gaussian. |

Table 5: Higher-order image moments.

| Parameter           | Description                                       |
|---------------------|---------------------------------------------------|
| num_islands         | Number of isolated pixel clusters after cleaning. |
| con_selected_island | Fraction of light in brightest island.            |
| con2                | Fraction of light in two brightest pixels.        |
| numb_pix            | Number of pixels passing cleaning.                |

Table 6: Image cleaning parameters.

| Parameter    | Description                                                                                |
|--------------|--------------------------------------------------------------------------------------------|
| tracking     | Telescope actively tracking (1) or not (0).                                                |
| good         | Telescope pointing accuracy flag (1 = good, 0 = not good).                                 |
| star         | Presence of bright star among cleaned pixels.                                              |
| edge         | Edge pixels passed cleaning.                                                               |
| weather_mark | Observation conditions related to atmospheric transparency (10 = excellent, -1 = no data). |
| alpha_c      | Camera rotation angle correction for orientation measurements.                             |

Table 7: Observation and telescope status flags.

| Parameter         | Description                                                               |
|-------------------|---------------------------------------------------------------------------|
| median_background | Median of pixel amplitudes that did not pass cleaning.                    |
| MAD_background    | Median absolute deviation of pixel amplitudes that did not pass cleaning. |

Table 8: Background estimation parameters.

| Parameter  | Description                                                             |
|------------|-------------------------------------------------------------------------|
| CR400phe   | Event count rate for size > 400 phe, averaged over 30 s.                |
| CR_portion | Average count rate for all events in a portion, not restricted by size. |

Table 9: Event-rate parameters.

# Appendix A: Co-ordinate Systems

## A.1 Horizontal Coordinate System

The horizontal coordinate system (also known as the Alt-Azimuth system) is a local coordinate system based on the observer's position on Earth. It is defined as follows:

1. **Reference Frame:** The system is centered on the observer, with the fundamental plane being the *horizon*.
2. **Coordinates:**
  - **Altitude (Alt or El):** The angle between the object and the horizon, measured from  $0^\circ$  (on the horizon) to  $90^\circ$  (at the zenith).
  - **Azimuth (Az):** The angle along the horizon, measured from the North point ( $0^\circ$ ), increasing towards the East up to  $360^\circ$ .
3. **Dependence on Observer's Location and Time:** Since it is based on the observer's horizon, the horizontal coordinates of an object change with both the observer's geographic location and the time of observation.

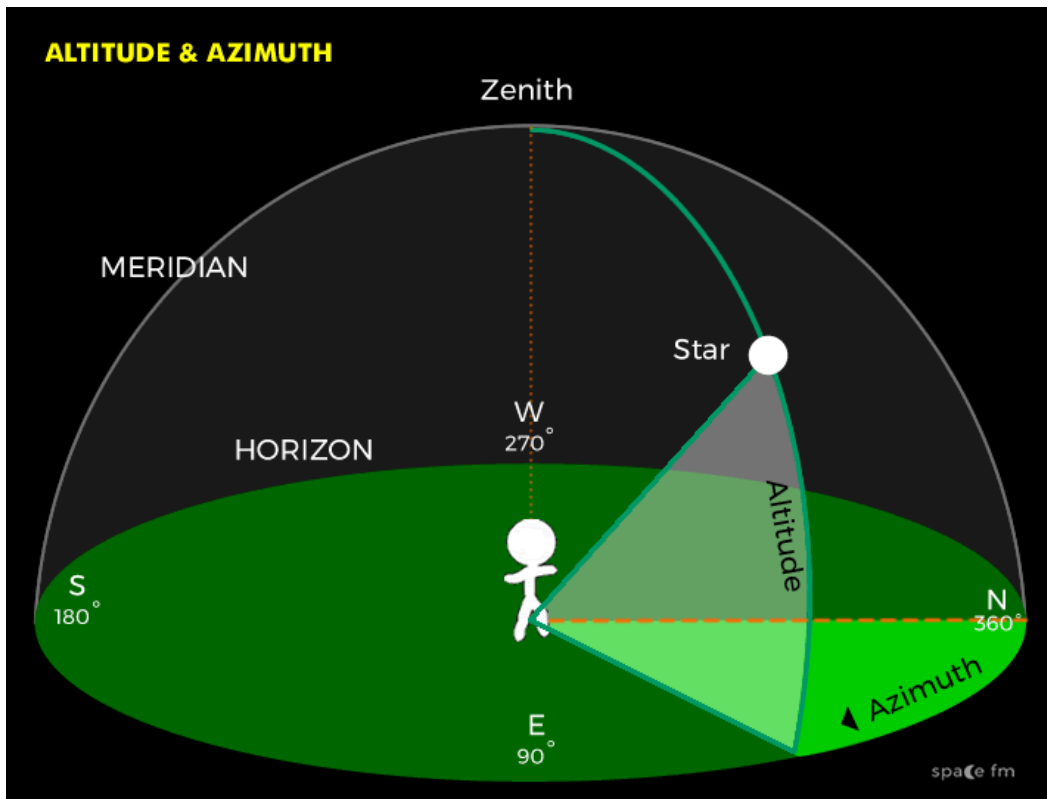


Figure 4: Horizontal Coordinate System

## A.2 Equatorial Coordinate System

The equatorial coordinate system is a celestial coordinate system that is independent of the observer's location and is fixed with respect to the stars. It is defined as follows:

1. **Reference Frame:** The system is based on the celestial sphere, with the fundamental plane being the *celestial equator*, which is the projection of Earth's equator onto the sky.
2. **Coordinates:**
  - **Right Ascension (RA):** The angular distance measured eastward along the celestial equator from the *vernal equinox*. It is usually expressed in hours, minutes, and seconds. It also can be measured in degrees.
  - **Declination (Dec):** The angular distance of the object north or south of the celestial equator, measured in degrees from  $-90^\circ$  (South Celestial Pole) to  $+90^\circ$  (North Celestial Pole).
3. **Independence from Location and Time:** Since it is related to the celestial sphere, the equatorial coordinates of a celestial object remain fixed (ignoring effects such as precession), regardless of the observer's location and time.

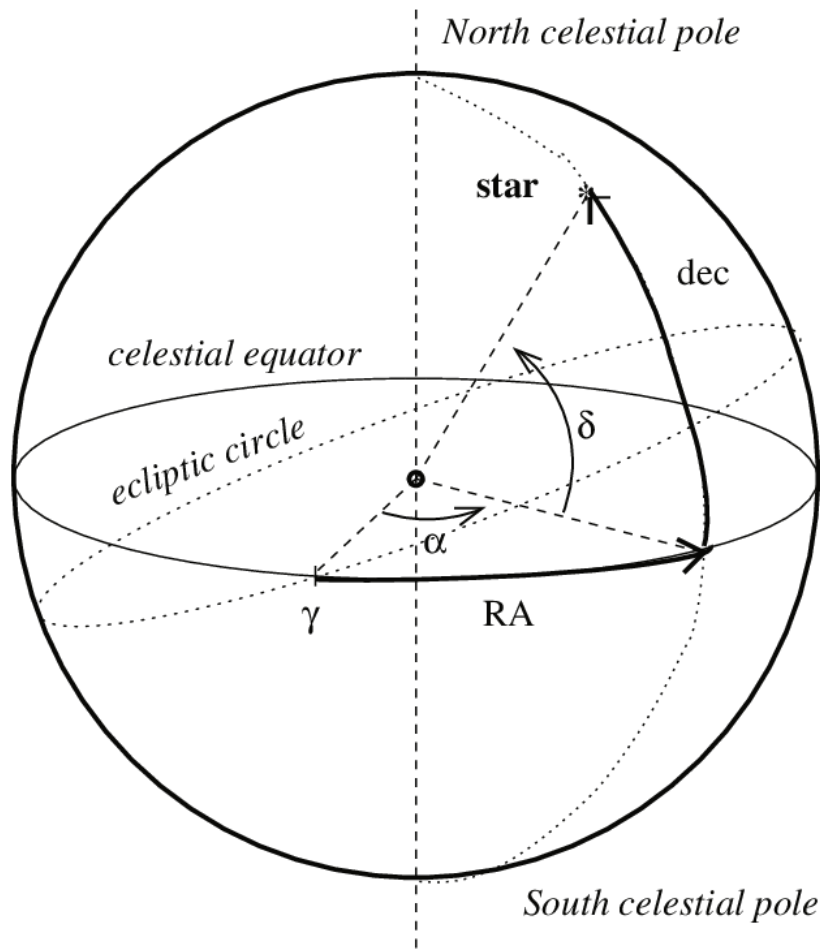


Figure 5: Equatorial Coordinate System

## Appendix B: Statistical Definitions

### 1. Mean: Central Tendency

- **Definition:** The mean represents the average value of a dataset, calculated by summing all data points and dividing by the number of observations.
- It provides a measure of the central location of the data.
- **Equation:** For a dataset  $x_1, x_2, \dots, x_n$ :

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i$$

### 2. Variance: Dispersion

- **Definition:** Variance quantifies the degree of spread or variability within a dataset. In simpler words, how much the data points deviate from the mean.
- In simpler words, how much the data points deviate from the mean. A higher variance indicates more spreading out.
- **Equation:**

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

## Distribution of Variance

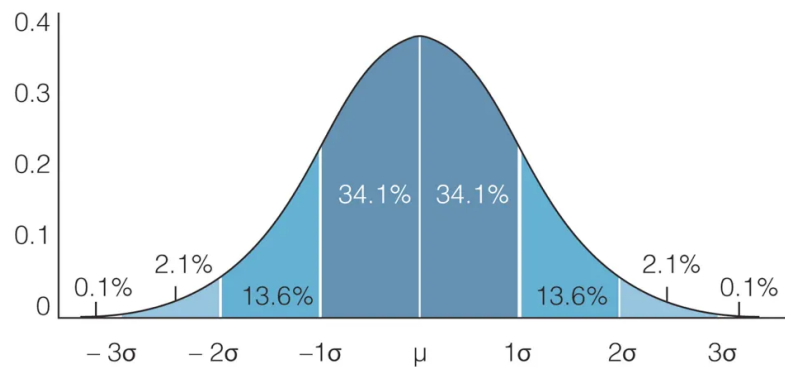


Figure 6: Distribution of Variance

### 3. Skewness: Asymmetry

- **Definition:** Skewness measures the asymmetry of the data distribution.
- **2 Types:**
  - Positive skewness: Longer right tail.

- Negative skewness: Longer left tail.

- **Equation:**

$$\text{Skewness} = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^3}{\left(\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2\right)^{3/2}}$$

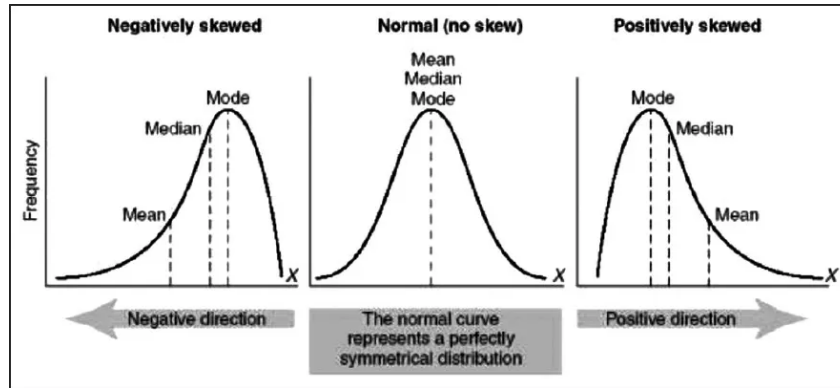


Figure 7: Skewness Shapes

#### 4. Kurtosis: Peakedness

- **Definition:** Kurtosis measures the "tailedness" or peakedness of the data distribution.

- **3 Types:**

- Leptokurtic: High peak and heavy tails.
- Mesokurtic: Normal peak and tails.
- Platykurtic: Low peak and light tails.

- **Equation:**

$$\text{Kurtosis} = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^4}{\left(\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2\right)^2}$$

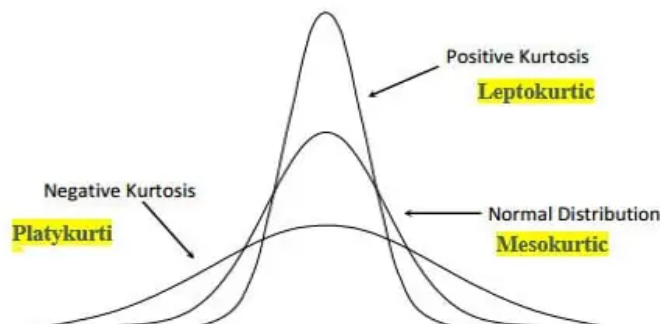


Figure 8: Kurtosis Types

These moments are interconnected. Mean and variance describe the central tendency and spread, while skewness and kurtosis provide additional information about the shape of the distribution.



## References

- [1] Justin Albert et al. “VHE gamma-ray observation of the Crab Nebula and its pulsar with the MAGIC telescope”. In: *The Astrophysical Journal* 674 (Dec. 2008), p. 1037. DOI: [10 . 1086 / 525270](https://doi.org/10.1086/525270).
- [2] A. M. Hillas. “Cerenkov Light Images of EAS Produced by Primary Gamma Rays and by Nuclei”. In: *19th International Cosmic Ray Conference (ICRC19), Volume 3*. Ed. by F. C. Jones. Vol. 3. International Cosmic Ray Conference. Aug. 1985, p. 445.